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SCIENCE

# COMPSCI 714
# Tutorial: Transfer Learning

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# Outline

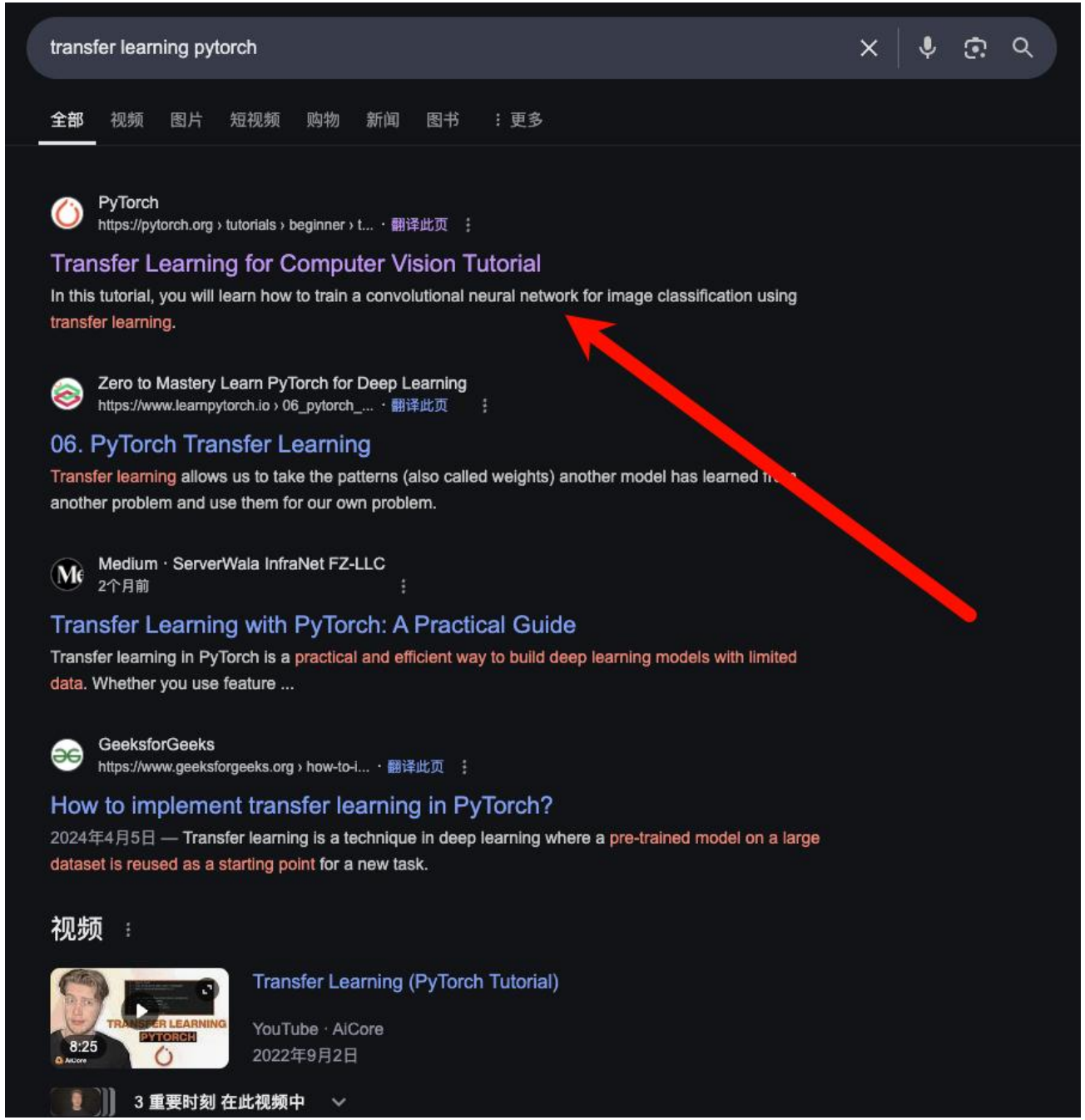
- Explore methods in transfer learning: 1. Fine-tuning 2. Model as feature extractor
- Challenge Problem

# Pre–request

- Google account
- Colab
- Google drive

# Instructions

Seach on Google



# Instructions

## Download data

### Load Data

We will use torchvision and torch.utils.data packages for loading the data.

The problem we're going to solve today is to train a model to classify **ants** and **bees**. We have about 120 training images each for ants and bees. There are 75 validation images for each class. Usually, this is a very small dataset to generalize upon, if trained from scratch. Since we are using transfer learning, we should be able to generalize reasonably well.

This dataset is a very small subset of imagenet.

**NOTE**

Download the data from here and extract it to the current directory.

```
# Data augmentation and normalization for training
# Just normalization for validation
data_transforms = {
    'train': transforms.Compose([
        transforms.RandomResizedCrop(224),
        transforms.RandomHorizontalFlip(),
        transforms.ToTensor(),
        transforms.Normalize([0.485, 0.456, 0.406], [0.229, 0.224, 0.225])
    ]),
    'val': transforms.Compose([
        transforms.Resize(256),
        transforms.CenterCrop(224),
        transforms.ToTensor(),
        transforms.Normalize([0.485, 0.456, 0.406], [0.229, 0.224, 0.225])
    ]),
}

data_dir = 'data/hymenoptera_data'
image_datasets = {x: datasets.ImageFolder(os.path.join(data_dir, x),
                                          data_transforms[x])
                  for x in ['train', 'val']}
```

# Instructions

Open it in Colab



## Transfer Learning for Computer Vision Tutorial

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**Author**: Sasank Chilamkurthy

In this tutorial, you will learn how to train a convolutional neural network for image classification using transfer learning. You can read more about the transfer learning at cs231n notes

Quoting these notes,

> In practice, very few people train an entire Convolutional Network from scratch (with random initialization), because it is relatively rare to have a dataset of sufficient size. Instead, it is common to pretrain a ConvNet on a very large dataset (e.g. ImageNet, which contains 1.2 million images with 1000 categories), and then use the ConvNet either as an initialization or a fixed feature extractor for the task of interest.

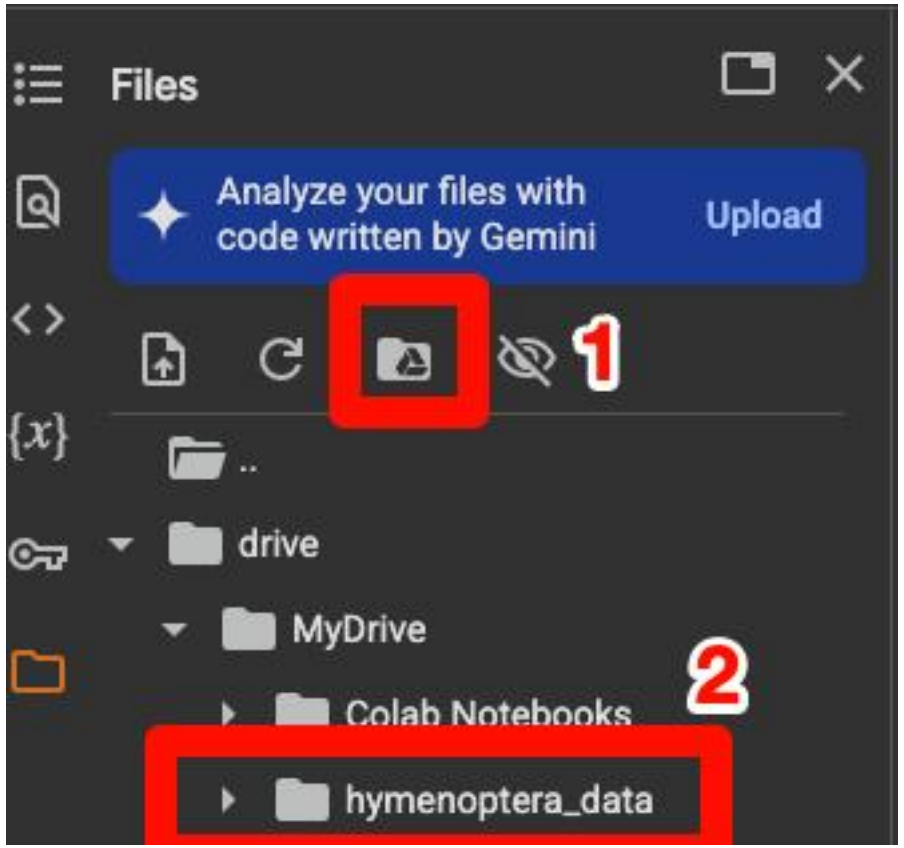
These two major transfer learning scenarios look as follows:

- **Finetuning the ConvNet**: Instead of random initialization, we initialize the network with a pretrained network, like the one that is trained on imagenet 1000 dataset. Rest of the training looks as usual.
- **ConvNet as fixed feature extractor**: Here, we will freeze the weights for all of the network except that of the final fully connected layer. This last fully connected layer is replaced with a new one with random weights and only this layer is trained.

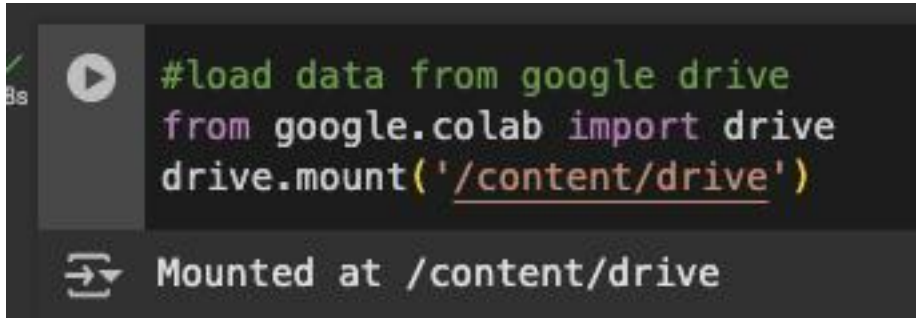
```
# License: BSD
# Author: Sasank Chilamkurthy

import torch
import torch.nn as nn
import torch.optim as optim
from torch.optim import lr_scheduler
import torch.backends.cudnn as cudnn
import numpy as np
import torchvision
from torchvision import datasets, models, transforms
import matplotlib.pyplot as plt
import time
import os
from PIL import Image
from tempfile import TemporaryDirectory

cudnn.benchmark = True
plt.ion()   # interactive mode
```

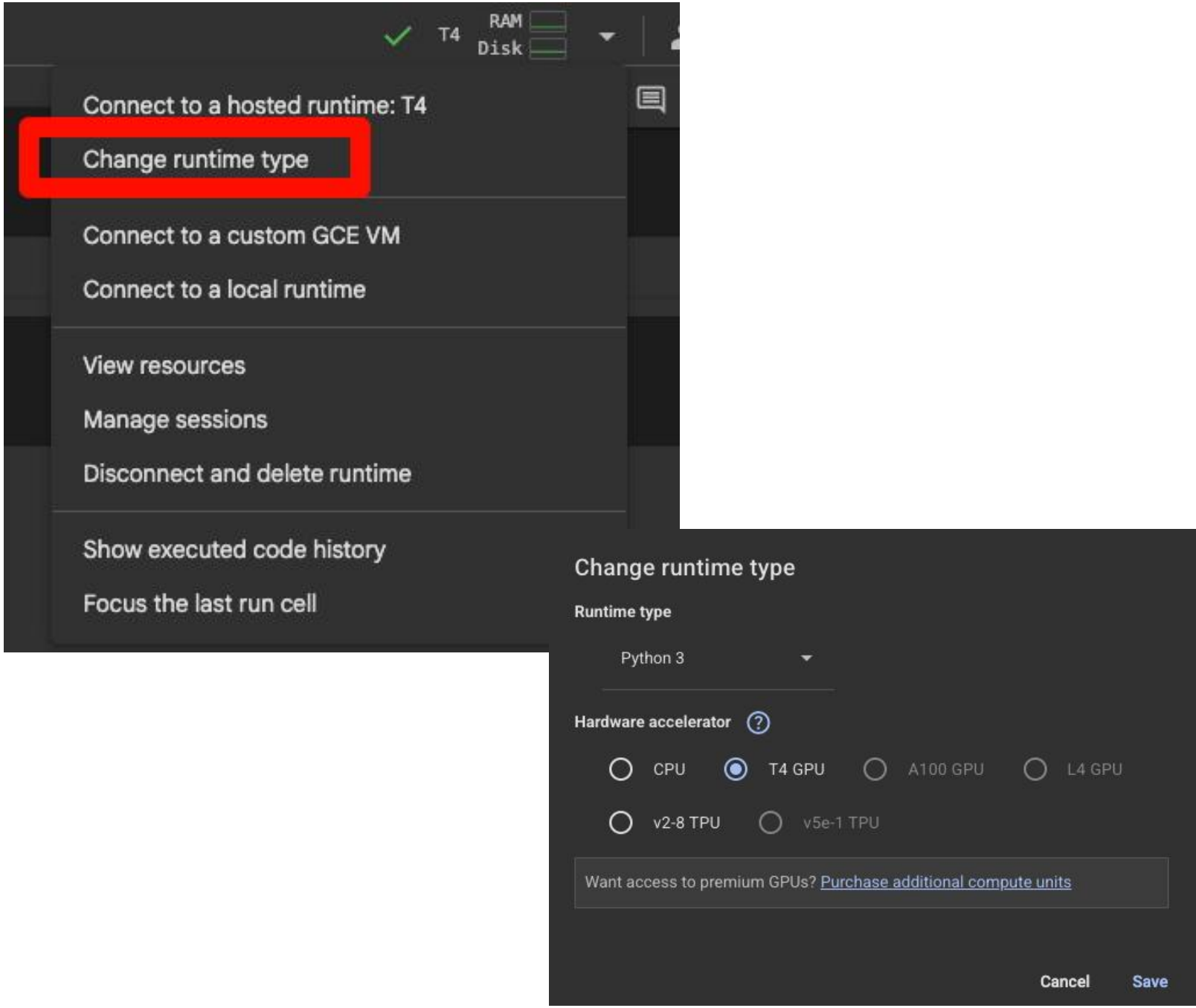
# Instructions

1. Connect to your Google drive
2. Upload the data




You should get this:



# Instructions

1. Change the runtime type
2. Choose T4 GPU